

A GOODNESS OF FIT TEST FOR A SCALE PARAMETER FAMILY OF DISTRIBUTIONS

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SUMMARY

Let G be the distribution function of a population; let F be a known distribution function and put $F_\theta(x) = F(x/\theta)$ where $\theta > 0$ is a scale parameter. Consider testing the hypothesis that $G = F_\theta$ for some unspecified θ . We propose a test statistic consisting of the ratio of two asymptotically efficient estimates of θ ; the asymptotic null distribution of this statistic is derived and tabled.

1. Introduction

Let $X_{1n}, \dots, X_{nn}$ be the order statistics of an independent sample from a population having distribution function G . Let F be a known distribution function and put

$$(1) \quad F_\theta(x) = F(x/\theta)$$

where $\theta > 0$ is a scale parameter. We consider testing the hypothesis $H_C : G = F_\theta$ for some unspecified θ .

Let $f(x) = F'(x)$ and assume that $f''(x)$ exists and $x^2 f''(x) \rightarrow 0$ as $x \rightarrow \pm \infty$. Define

$$(2) \quad H(u) = F^{-1}(u), \quad L(y) = -1 - yf'(y)/f(y), \quad J(u) = 1 - L'(H(u))$$

where

$$(3) \quad I = \int_{-\infty}^{\infty} L^2(y)f(y)dy = \int_0^1 L(H(u))^2 du = \int_0^1 L'(H(u))H(u)du.$$

In what follows we abbreviate $c(i/(n+1))$ by c_{in} for functions $c(u)$ defined for $u \in (0, 1)$, $i = 1, \dots, n$. The motivation for the test to be defined and discussed below is heuristic in the sense that the various statements are valid only under certain regularity conditions; these are concerned with the validity of Cramér-Rao bounds, the asymptotic normality of

functions of order statistics and the validity of various integration operations (see e.g. Chernoff, Gastwirth and Johns (1967)); these conditions will not be spelled out explicitly here as it is simpler to verify the various statements directly in applications where one deals with a specific F .

As test statistic for HC, we propose

$$(4) \quad S_n = S_{1n}/S_{2n},$$

where

$$S_{1n} = n^{-1} \sum_{i=1}^n J(i/(n+1))X_{in}, S_{2n} = (n^{-1} \sum_{i=1}^n J_{in} X_{in}^2 / H_{in})^{1/2}.$$

Under HC both S_{1n} and S_{2n} are asymptotically efficient estimates of θ (subject to some provisos – see Chernoff, Gastwirth and Johns (1967) e.g.); hence, under HC, generally, $S_n = 1 + o_p(1)$.

When HC is not necessarily true, we generally get

$$S_n = \int_0^1 J(u)G^{-1}(u)du / \left(\int_0^1 J(u)G^{-1}(u)^2 du / H(u) \right)^{1/2} + o_p(1).$$

The numerator of this ratio is

$$\int_0^1 (J(u)H(u))^{1/2} (J(u)G^{-1}(u)^2 / H(u))^{1/2} du$$

so that by Schwarz's inequality the ratio is in absolute value no more than $\int_0^1 J(u)H(u)du = 1$ with equality only if there exists constants a and b such that $a(J(u)H(u))^{1/2} + b(J(u)G^{-1}(u)^2 / H(u))^{1/2} = 0$ for a.a. u which implies that $G^{-1}(u) = \theta H(u)$ for some θ i.e. the truth of HC. Consequently it is reasonable to reject HC when $S_n \leq 1 - \epsilon_n$ for ϵ_n suitably chosen and in general these tests should be consistent.

Another way in which S_n can be motivated is the following:

Suppose we wish to test $H_S : G = F_{\theta_0}$ with θ_0 specified – then under H_S , EX_{in}/θ_0 is approximately $\theta_0^{-1} F_{\theta_0}^{-1}(i/(n+1)) = F^{-1}(i/(n+1)) = H_{in}$, and a plausible class of test statistics for H_S is

$$\Sigma (X_{in}/\theta_0 - H_{in})^2 W(i/(n+1))$$

with W a suitable weight function. One possible choice for W is $W(u) = J(u)/H(u)$ resulting in the statistic

$$(7) \quad E_n^0 = \Sigma (X_{in}/\theta_0 - H_{in})^2 J_{in}/H_{in}.$$

Now, in case the hypothesis does not specify θ_0 we can think of using the statistic

$$(8) \quad E_n = \Sigma (X_{in}/S_{1n} - H_{in})^2 J_{in}/H_{in}$$

in which θ_0 is estimated by S_{1n} ; squaring out E_n we get

$$(9) \quad E_n = nS_n^{-2} - 2n + \Sigma J_{in} H_{in}$$

so that E_n and S_n are equivalent statistics for HC.

In paragraph 2 we outline a method by means of which the asymptotic null distribution of E_n^0 can be derived and in paragraph 3 the same is done for E_n and S_n . In paragraphs 4 and 5 we specialize to the gamma family of distributions obtaining more explicit results and tabling the distribution and its percentiles.

2. The asymptotic null distribution of E_n^0

Suppose HS is true; we may assume $\theta_0 = 1$ without loss of generality. Let $U_{in} = F(X_{in})$ and $V_{in} = U_{in} - (i/(n+1))$. Then, by Taylor's theorem

$$(10) \quad X_{in} = H_{in} + V_{in} H'_{in} + \frac{1}{2} V_{in}^2 H''_{in}(U_{in}^*)$$

with U_{in}^* between U_{in} and $i/(n+1)$.

Substituting into (7) we get

$$(11) \quad E_n^0 = Q_n + \frac{1}{4} R_{1n} + R_{2n}$$

with

$$(12) \quad Q_n = \Sigma V_{in}^2 H_{in}'^2 J_{in}/H_{in}$$

$$(13) \quad R_{1n} = \Sigma V_{in}^4 H_{in}''(U_{in}^*)^2 J_{in}/H_{in}$$

$$(14) \quad R_{2n} = \Sigma V_{in}^3 H_{in}' H_{in}''(U_{in}^*) J_{in}/H_{in}.$$

Using the technique of De Wet and Venter (1972b) we find that

$$Q_n = (1 + O_p(n^{-1/2})) T_{n+1}^0$$

with

$$T_{n+1}^0 = \sum_{i,j} c_{ijn+1}^0 Z_i Z_j$$

$$c_{ijn+1}^0 = (n+1)^{-2} \sum_k \psi_{ikn} \psi_{jkn} H_{kn}'^2 J_{kn} / H_{kn}$$

$$\psi(x, y) = \begin{cases} 1-y & \text{for } x \leq y \\ -y & \text{for } x > y \end{cases}$$

$$\psi_{ijn} = \psi(i/(n+1), j/(n+1)),$$

and $\{Z_i + 1\}$ independent random variables each exponentially distributed with expectation one.

To get the asymptotic distribution of T_n^0 we can proceed as in paragraphs 2 and 3 of De Wet and Venter (1972a). Setting

$$(15) \quad c^0(x, y) = \int_0^1 \psi(x, u) \psi(y, u) H'(u)^2 J(u) du / H(u),$$

the problem reduces to that of finding the eigenvalues $\{\gamma_m\}$ and eigenfunctions of c^0 (if they exist) and to verify the relevant conditions to arrive at the result that

$$(16) \quad D(Q_n - e_n^0) \rightarrow D(\sum_{m=1}^{\infty} \gamma_m (Y_m^2 - 1))$$

with $Y_1, Y_2, \dots$ independent $N(0, 1)$ random variables and

$$e_n^0 = (n+1)^{-1} \sum_{k=1}^n (k/(n+1))(1-k/(n+1)) H_{kn}'^2 J_{kn} / H_{kn}.$$

Finally if it can be shown that $R_{1n}, R_{2n} = o_p(1)$, it then follows that $E_n^0 - e_n^0$ has the asymptotic distribution shown in (16).

3. The asymptotic null distributions of S_n and E_n

Supposing now that HC is true we may take it that $\theta = 1$, as both S_n and E_n are scale invariant.

Then we have

$$\begin{aligned} (17) \quad S_{2n}^2 (1 - S_n^2) &= S_{2n}^2 - S_{1n}^2 \\ &= n^{-1} \sum J_{in} X_{in}^2 / H_{in} - (n^{-1} \sum J_{in} X_{in})^2. \end{aligned}$$

Substituting from (10) we find

$$nS_{2n}^2 (1 - S_n^2) = Q_n - ny_n^2 + a_n + R_{3n}$$

with Q_n as above

$$y_n = n^{-1} \sum V_{in} H'_{in} J_{in}$$

$$a_n = \sum J_{in} H_{in} (1 - n^{-1} \sum J_{in} H_{in}).$$

$$\begin{aligned} R_{3n} = & \sum J_{in} e_{in}^2 / H_{in} + 2 \sum J_{in} e_{in} (1 - n^{-1} \sum J_{in} H_{in}) + 2 \sum J_{in} H_{in} e_{in} V_{in} / H_{in} \\ & - n^{-1} (\sum J_{in} e_{in})^2 - 2n^{-1} (\sum J_{in} e_{in}) (\sum J_{in} H'_{in} V_{in}) \\ & + 2(1 - n^{-1} \sum J_{in} H_{in}) (\sum J_{in} H'_{in} V_{in}) \end{aligned}$$

with $e_{in} = \frac{1}{2} V_{in}^2 H''(U_{in}^*)$.

Using the same technique referred to above, we get

$$Q_n - ny_n^2 = (1 + O_p(n^{-1/2})) T_{n+1}$$

with

$$T_{n+1} = \sum_{ij} c_{ijn+1} Z_i Z_j \text{ with } c_{ijn} = c_{ijn}^0 - I^{-1} c_{in}^1 c_{jn}^1$$

with c_{ijn}^0 as above and

$$c_{in}^1 = I^{-1/2} n^{-1/2} (n+1)^{-1} \sum_k \psi_{ikn} H'_{kn} J_{kn}.$$

In order to get the asymptotic distribution of T_n we must therefore find the eigenvalues and eigenfunctions of

$$c(x, y) = c^0(x, y) - I^{-1} c^1(x) c^1(y),$$

with c^0 as above and

$$c^1(x) = I^{-1/2} \int_0^1 \psi(x, u) H'(u) J(u) du.$$

We now show that under reasonable conditions c^1 is an eigenfunction of c^0 with corresponding eigenvalue I^{-1} .

Substituting for ψ and integrating partially we find that

$$c^1(x) = I^{-1/2} L(H(x)),$$

if $uL(H(u)) \rightarrow 0$ as $u \rightarrow 0$, $(1-u)L(H(u)) \rightarrow 1$ and $yf(y) \rightarrow 0$ as $y \rightarrow \pm \infty$. Thus, if changing of the order of integration is permitted,

$$\begin{aligned} \int_0^1 c^0(x, y) c^1(y) dy &= -I^{-1/2} \int_0^1 \psi(x, u) H'(u)^2 J(u) / H(u) \left(\int_0^1 \psi(y, u) L(H(y)) dy \right) du \\ &= -I^{-1/2} \int_0^1 \psi(x, u) H'(u)^2 J(u) / H(u) \left(\int_0^u L(H(y)) dy \right) du. \end{aligned}$$

Substituting for L and integrating $\int_0^u L(H(y)) dy$ becomes $-H(u)f(H(u))$.

It follows that $\int_0^1 c^0(x, y) c^1(y) dy = I^{-1} c^1(x)$.

Consequently, if the eigenvalues of c^0 are $\gamma_1 = I^{-1}$, $\gamma_2, \dots$ it follows that the eigenvalues of c are $\gamma_2, \gamma_3, \dots$ and if the conditions of theorems 1 or 2 of De Wet and Venter (1972a) can be shown to be satisfied we would be lead to conclude that

$$D(Q_n - n\gamma_n^2 - e_n) \rightarrow D(\Sigma_2^\infty \gamma_m(Y_m^2 - 1))$$

with $e_n = e_n^0 - e_n^1$ where e_n^0 is as above and

$$e_n^1 = n^{-1} (n+1)^{-1} \sum_{i,j=1}^n H'_{in} H'_{jn} J_{in} J_{jn} T_{ijn}$$

where $T(x, y) = \begin{cases} x(1-y) & \text{for } x \leq y \\ y(1-x) & \text{for } x > y \end{cases}$

and $T_{ijn} = T(i/(n+1), j/(n+1))$.

If it can be shown that $R_{3n} = o_p(1)$ then we have that

$$(18) \quad D(nS_{2n}^2 (1 - S_n^2) - a_n - e_n) \rightarrow D(\Sigma_2^\infty \gamma_m(Y_m^2 - 1)).$$

Now, $S_{2n}^2 = 1 + o_p(n^{-1/2})$; hence if $a_n, e_n = o(n^{1/2})$ then we would have

$$D(n(1 - S_n^2) - a_n - e_n) \rightarrow D(\Sigma_2^\infty \gamma_m(Y_m^2 - 1)).$$

Finally, seeing that $1 - S_n^2 = (1 - S_n)(1 + S_n)$ with $S_n = 1 + o_p(1)$ it then follows that $S_n = 1 + o_p(n^{-1/2})$ and hence that also

$$D(2n(1 - S_n) - a_n - e_n) \rightarrow D(\Sigma_2^\infty \gamma_m (Y_m^2 - 1)).$$

From (9), $S_{In}^2(E_n - n(n^{-1} \Sigma J_{in} H_{in} - 1)) = n(S_{2n}^2 - S_{In}^2)$.

Comparing this with (17) and (18) we then find that $S_{In}^2(E_n - n(n^{-1} \Sigma J_{in} H_{in} - 1) - a_n - e_n) = S_{In}^2(E_n - e_n + n(n^{-1} \Sigma J_{in} H_{in} - 1)^2)$ has the asymptotic distribution in (18). Finally, if $S_{In}^2 = 1 + o_p(n^{-1/2})$ and $n(n^{-1} \Sigma J_{in} H_{in} - 1)^2 = o(1)$, this is also the asymptotic distribution of $E_n - e_n$.

Remark

Comparing the asymptotic distributions of E_n^0 and E_n we note the absence of the term $\gamma_1(Y_1^2 - 1)$ in the latter; this phenomenon is similar to that of a loss of one degree of freedom associated with a χ^2 -test for HC as compared with a χ^2 -test of HS.

4. The asymptotic distributions of E_n^0 and E_n for the gamma family of distributions

In this paragraph we take F to be the gamma distribution function with parameter r i.e.

$$F(x) = \int_0^x y^{r-1} e^{-y} dy / \Gamma(r),$$

where we assume r known. Since the work in this paragraph follows closely that in De Wet and Venter (1972b), we will be quite brief.

Before deriving the asymptotic distribution of E_n^0 we give two preliminary results.

Lemma 1

If $t_n = n^{-1} \Sigma H_{in} - r$, then $t_n = O(n^{-1} \log n)$ as $n \rightarrow \infty$.

Proof:

This follows in the same way as lemma 2 of De Wet and Venter (1972b).

Lemma 2

Let U_{in} and U_{in}^* be as before and set

$$R_n(\alpha, \beta, \gamma, \tau) = \Sigma V_{in}^\alpha (H'_{in})^\beta H''(U_{in}^*)^\gamma / H_{in}^\tau$$

Then $R_n(\alpha, \beta, \gamma, \tau)$ is:

(1) $o_p(1)$ if $\alpha=4, \beta=0, \gamma=2, \tau=1$ or $\alpha=3; \beta=\gamma=\tau=1$.

(2) $o_p(n^{1/2} \log n)$ if $\alpha=\beta=1$ and $\gamma=\tau=0$.

(3) $o_p((\log n)^2)$ if $\alpha=2, \gamma=1$ and $\beta=\tau=0$.

Proof:

Since the proof is essentially the same as that of lemma 4 of De Wet and Venter (1972b) it will be omitted. Details can be found in De Wet and Venter (1972c).

We now turn to the asymptotic distribution of E_n^O . Note that in this case $J(u) = r^{-1}$.

Theorem 1

Under HS we have

$$D(r(E_n^O - e_n^O)) \rightarrow D(\Sigma(Y_m^2 - 1)/m)$$

where $Y_1, Y_2, \dots$ are independent $N(0,1)$ random variables.

Proof:

In the notation of lemma 2 we have $R_{1n} = r^{-1}R_n(4, 0, 2, 1) = o_p(1)$ and $R_{2n} = r^{-1}R_n(3, 1, 1, 1) = o_p(1)$. Hence $E_n^O = Q_n + o_p(1)$. In De Wet and Venter (1972a) it is shown that $D(r(Q_n - EQ_n)) \rightarrow D(\Sigma_1^\infty(Y_m^2 - 1)/m)$. Since $EQ_n = (n+1)/(n+2)e_n^O$ the theorem follows.

Remark

Note that the asymptotic distribution of $r(E_n^O - e_n^O)$ is independent of r and is the same as that of $L_n^O - a_n^O$ used in De Wet and Venter (1972b) as a test statistic for normality.

Theorem 2

Under HC we have

$$D(r(E_n - e_n)) \rightarrow D(\Sigma_2^\infty(Y_m^2 - 1)/m)$$

$$D(r(2n(1 - S_n) - a_n - e_n)) \rightarrow D(\Sigma_2^\infty(Y_m^2 - 1)/m).$$

Proof

Using lemmas 1 and 2 it follows that $R_{3n} = o_p(1)$. It can be shown that the conditions of theorem 1 of De Wet and Venter (1972a) are satisfied. (See De Wet and Venter (1972c) for details).

It can also be shown that $e_n = o(\log n)$. From this the theorem follows.

Remark

1. Note again the lack of dependence on r of the asymptotic distribution.
2. For the special case $r = 1$ (i.e. testing of exponentiality) many tests have been proposed; the tests discussed here resemble that of Jackson (1967) in that a ratio of two combinations of the order statistics is also used there. However, the numerator of his test statistic is not an efficient estimate of the parameter of which the denominator is an efficient estimate.

5. **Inversion of characteristic functions**

Since the asymptotic distribution of $r(E_n^0 - e_n^0)$ has already been tabled in De Wet and Venter (1972b), we will confine ourselves to tabling the distribution function of $\Sigma_2^\infty (Y_m^2 - 1)/m$.

Let Ψ denote this distribution function and f the corresponding characteristic function. It then follows that $\hat{f}(t) = r(t)e^{i\theta(t)}$ with

$$(19) \quad r(t) = \prod_{k=2}^{\infty} (1 + 4t^2/k^2)^{-1/4}$$

$$r(t) = (e^{2\pi t} - e^{-2\pi t/4\pi t})^{-1/4} (1 + 4t^2)^{1/4}$$

and

$$(20) \quad \theta(t) = \frac{1}{2} \sum_{k=2}^{\infty} (\tan^{-1} 2t/k - 2t/k).$$

Using the inversion formula on p. 483 of Feller (1966), it follows that

$$(21) \quad \Psi(y) - \Psi(0) = 1/2\pi \int_{-\infty}^{\infty} (f(t)/it)(1 - e^{-ity})dt.$$

Substituting for $\hat{f}(t)$ in (21) and taking real parts, yield

$$(22) \quad \Psi(y) - \Psi(0) = 1/\pi \int_0^{\infty} (r(t)/t)(\sin \theta(t) + \sin(ty - \theta(t)))dt.$$

The integration in (22) can be carried out numerically. Taking y very large in (22)

we can obtain $1 - \Psi(0)$ and thus $\Psi(y)$ for all y . This distribution function is tabulated in table 1. Let $y(\alpha)$ be such that $\Psi(y(\alpha)) = 1 - \alpha$. Table 2 gives $y(\alpha)$ for certain choices of α . Tables 3 and 4 gives the constants e_n^0, e_n and a_n for $r = 1, 2, 3, 4, 5$ and certain choices of n . From this it follows that $e_n^0 \cong e_n - r^{-1}$. This also follows from the fact that

$$e_n \rightarrow r^{-2} \int_0^1 \int_0^1 H'(x)H'(y)T(x, y)dxdy = r^{-2}r = r^{-1}.$$

TABLE 1.

The distribution function $\Psi(y)$

y	$\Psi(y)$	y	$\Psi(y)$	y	$\Psi(y)$	y	$\Psi(y)$
-3,0	0,000	-0,9	0,212	0,8	0,796	2,5	0,969
-2,5	0,001	-0,8	0,250	0,9	0,816	2,6	0,972
-2,4	0,001	-0,7	0,289	1,0	0,834	2,7	0,975
-2,3	0,001	-0,6	0,329	1,1	0,851	2,8	0,978
-2,2	0,002	-0,5	0,370	1,2	0,866	2,9	0,980
-2,1	0,004	-0,4	0,411	1,3	0,880	3,0	0,982
-2,0	0,007	-0,3	0,451	1,4	0,892	3,1	0,984
-1,9	0,011	-0,2	0,491	1,5	0,903	3,2	0,986
-1,8	0,017	-0,1	0,530	1,6	0,913	3,3	0,987
-1,7	0,025	-0,0	0,567	1,7	0,922	3,4	0,989
-1,6	0,036	0,1	0,602	1,8	0,931	3,5	0,990
-1,5	0,051	0,2	0,636	1,9	0,938	4,0	0,995
-1,4	0,069	0,3	0,668	2,0	0,944	4,5	0,997
-1,3	0,091	0,4	0,697	2,1	0,951	5,0	0,999
-1,2	0,116	0,5	0,725	2,2	0,956	6,0	1,000
-1,1	0,145	0,6	0,751	2,3	0,960		
-1,0	0,177	0,7	0,774	2,4	0,965		

TABLE 2.

The percentiles of ψ

α	$y(\alpha)$
0,20	0,8186
0,15	1,0943
0,10	1,4725
0,05	2,0950
0,01	3,4960

TABLE 3.

The constants e_n^0 and e_n

r										
n	1		2		3		4		5	
	e_n^0	e_n	e_n^0	e_n	e_n^0	e_n	e_n^0	e_n	e_n^0	e_n
1	0,72	0,22	0,38	0,06	0,26	0,03	0,19	0,02	0,16	0,01
5	1,40	0,69	0,75	0,34	0,51	0,23	0,38	0,17	0,31	0,13
10	1,64	0,85	0,88	0,45	0,60	0,30	0,45	0,23	0,36	0,18
20	1,85	0,99	1,00	0,54	0,68	0,37	0,51	0,28	0,41	0,22
30	1,96	1,06	1,05	0,58	0,72	0,40	0,55	0,31	0,44	0,25
40	2,03	1,11	1,09	0,62	0,74	0,42	0,57	0,32	0,45	0,26
50	2,08	1,15	1,12	0,64	0,76	0,44	0,58	0,34	0,47	0,27
100	2,22	1,27	1,19	0,71	0,81	0,49	0,62	0,37	0,50	0,30
150	2,28	1,32	1,23	0,74	0,84	0,51	0,64	0,39	0,52	0,32
200	2,35	1,37	1,25	0,76	0,86	0,53	0,66	0,41	0,53	0,33
400	2,46	1,48	1,31	0,82	0,90	0,57	0,69	0,44	0,55	0,35
600	2,52	1,53	1,34	0,84	0,92	0,59	0,70	0,45	0,57	0,37
800	2,56	1,57	1,36	0,86	0,93	0,60	0,71	0,46	0,57	0,37
1000	2,59	1,60	1,38	0,88	0,94	0,61	0,72	0,47	0,58	0,38

TABLE 4.

The constant a_n

r

n	1	2	3	4	5
1	0,21	0,13	0,10	0,08	0,06
5	0,69	0,40	0,28	0,21	0,17
10	1,00	0,56	0,39	0,30	0,24
20	1,34	0,74	0,51	0,39	0,31
30	1,55	0,85	0,59	0,45	0,36
40	1,70	0,93	0,64	0,49	0,39
50	1,82	0,99	0,68	0,52	0,42
100	2,18	1,18	0,81	0,62	0,50
150	2,39	1,29	0,89	0,68	0,55
200	2,54	1,37	0,94	0,72	0,58
400	2,89	1,56	1,07	0,82	0,66
600	3,10	1,67	1,15	0,88	0,71
800	3,25	1,75	1,20	0,92	0,74
1000	3,36	1,81	1,25	0,95	0,77

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